

More Measures of Vulnerability: Splinter Sets and Directed Toughness

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ABSTRACT. We introduce the splinter set of a graph which records all triples (s, m, k) such that there exists a set S of s vertices whose removal leaves k components with maximum component order m . From this set several parameters can be calculated such as connectivity, toughness and integrity. For a parameter ψ that can be calculated from the splinter set, we define ψ for directed graphs by replacing component by strong component in the definition, and define the directed value $\bar{\psi}$ for an undirected graph as the maximum value of ψ over all orientations. As an example we explore the properties of directed toughness.

1 Introduction

Many graphical parameters are concerned about numbers of components and how hard or easy it is to create them. The obvious one is connectivity, but there is also toughness for example. If one also considers the maximum order of a component, then examples include vertex cover number, integrity and tenacity.

In this paper we define the splinter set of a graph which records the disruptive power of all sets of vertices in terms of the cardinality of the set and the number of components and maximum component order of what remains when the set is removed. We observe that from this set several parameters can be deduced including the ones mentioned above.

In another direction, we observe that when a measure of vulnerability is determined by the splinter set, a parameter for digraphs can be obtained by replacing “component” by “strong component” in the definition. This leads to directed versions of connectivity, toughness etc. Following the lead of [12], we also define the “directed” measure for undirected graphs as the maximum value of the parameter over all orientations. As an example we explore the directed toughness of a graph.

2 The Splinter Set

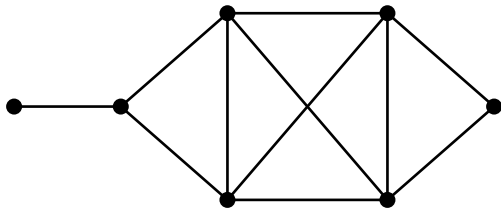
For a graph H , we let $m(H)$ denote the maximum order of a component of H , and we let $k(H)$ denote the number of components of H .

Given a graph G , we define the *(vertex) splinter set* $\mathcal{S}(G)$ of G as the set of all triples

$$(|S|, m(G - S), k(G - S))$$

where S ranges over all subsets of the vertex set. (We define it to be a set, but it is feasible that in some applications a multiset would be better.)

Consider for example the following graph: Apart from triples of the form $(s, 7-s, 1)$ the triples in the splinter set are: $(1, 5, 2)$, $(2, 3, 2)$, $(2, 4, 2)$, $(3, 3, 2)$, $(3, 2, 2)$, $(3, 2, 3)$, $(4, 2, 2)$, $(4, 1, 3)$ and $(5, 1, 2)$.



In the definitions of vulnerability, where the worst S is to be considered, it suffices to consider the “minimal” elements: (s small as possible, m as small as possible, k as large as possible). So we say that $(s, m, k) \in \mathcal{S}$ is minimal if there is no distinct $(s', m', k') \in \mathcal{S}$ with $s' \leq s$, $m' \leq m$ and $k' \geq k$. For the above examples, the minimal triples are $(0, 7, 1)$, $(1, 5, 2)$, $(2, 3, 2)$, $(3, 2, 3)$ and $(4, 1, 3)$.

The key idea is that $\mathcal{S}(G)$ contains in it considerable information about the graph. For example:

- connectivity: $\kappa(G) = \min\{s : (s, m, k) \in \mathcal{S}, k \geq 2\}$
- l -connectivity: $\kappa_l(G) = \min\{s : (s, m, k) \in \mathcal{S}, k \geq l\}$
- vertex cover: $\alpha(G) = \min\{s : (s, m, k) \in \mathcal{S}, m = 1\}$
- integrity: $I(G) = \min\{s + m : (s, m, k) \in \mathcal{S}\}$
- toughness: $t(G) = \min\{s/k : (s, m, k) \in \mathcal{S}, k \geq 2\}$
- tenacity: $t(G) = \min\{(s + m)/k : (s, m, k) \in \mathcal{S}, k \geq 2\}$
- component order connectivity: $\kappa_c^{(a)} = \min\{s : (s, m, k) \in \mathcal{S}, m < a\}$

Integrity was introduced in [1], toughness by [4] and tenacity in [5]. Component order connectivity is studied in [3] inter alia (and is related to the vector D_i defined in [8]). It follows that finding the splinter set is NP-complete.

For digraphs, the definition of the *(vertex) splinter set* is the same except that we replace “component” by “strong component” in the definition. Then:

- strong-connectivity: $\kappa(D) = \min\{s : (s, m, k) \in \mathcal{S}, k \geq 2\}$
- decycling number: $\nabla(D) = \min\{s : (s, m, k) \in \mathcal{S}, m = 1\}$
- integrity: $I(D) = \min\{s + m : (s, m, k) \in \mathcal{S}\}$
- toughness: $t(D) = \min\{s/k : (s, m, k) \in \mathcal{S}, k \geq 2\}$

The decycling number is studied in [2] and elsewhere, and the integrity is introduced in [1]. We introduce the notion of the toughness of a directed graph.

Edge splinter sets and edge versions of the above can easily be defined.

3 Directed Measures for Undirected Graphs

Our main interest is undirected graphs. We observe that for the above parameters, in fact for any graphical parameter ψ on digraphs, one can define for undirected graph G :

$$\bar{\psi}(G) = \max\{\psi(D) : D \text{ is orientation of } G\}$$

If G is a forest, then any orientation of G has each strong component order 1. So forests typically have the extreme value of $\bar{\psi}(G)$.

Example. Integrity is studied in [12].

Example. Edge connectivity. For the edge connectivity λ , there is a simple formula.

Lemma 1 *For all graphs G*

$$\bar{\lambda}(G) = \lfloor \lambda(G)/2 \rfloor.$$

PROOF. The RHS value is obviously an upper bound, as one considers a minimum edge cut T of G . Then in any orientation D of G , pick the direction with fewest arcs: the removal of that disconnects D .

On the other hand, Nash-Williams [9] (extending Robbin's theorem) showed that every $2k$ -edge-connected graph has a k -strong-edge-connected orientation. (There is a nice discussion in [13].) QED

For connectivity it is not as clear. By Robbin's theorem, a 2-connected graph has a strongly-connected orientation. However, some 2-connected graphs have 2-strongly-connected orientations and others not. For the latter consider the cycle. For the former, start with the bowtie $K_1 + 2K_2$ and orient it to be strongly connected. Then expand the bowties by duplicating the central vertex once, and duplicating the other four vertices each m times. The minimum cut-set is both central vertices.

4 Directed Toughness

For an overview of the elementary properties of toughness, see [7]. We start with some bounds on $\bar{t}$. Note that (from Lemma 1, for example):

$$\bar{\kappa}(G) \leq \delta(G)/2.$$

Then

Lemma 2 *For any graph G of order p ,*

- (a) $\bar{t}(G) \leq \bar{\kappa}(G)/2$;
- (b) $\bar{t}(G) \leq \nabla(G)/(p - \nabla(G))$;
- (c) *if G is 2-edge-connected, then $\bar{t} \geq 1/(p - 1)$.*

PROOF. (a) Removing any cut-set produces at least two components.

(b) Removing a minimum decycling set yields $\nabla(G)$ components.

(c) By Robbins' theorem, there is always a connected orientation. QED

An immediate corollary of parts (b) and (c) is:

Lemma 3 *For the cycle C_p on p vertices, $\bar{t}(C_p) = 1/(p - 1)$.*

4.1 Cliques

The small cliques have directed toughness as follows: $\bar{t}(K_3) = 1/2$ and $\bar{t}(K_4) = 1/3$. Note that this shows that directed toughness is not monotone. For the complete graph in general, the following result gives the asymptotics:

Theorem 1 *The directed toughness of the complete graph K_n is $O(n/\log n)$.*

PROOF. Reid and Parker [11] showed that for any tournament of order n there is always a transitive tournament of order at least $O(\log n)$. By letting S be the vertices outside the transitive tournament, it follows that for any tournament T on n vertices $\bar{t}(T) \leq O(n/\log n)$.

To prove the lower bound, we show that the random tournament T has $\bar{t}(T) \geq O(n/\log n)$. For this it suffices to show (a) the maximum cardinality of a transitive subgraph is $O(\log n)$; and (b) the connectivity is at least εn . For, it then follows that every cut set S has linear size, and its removal leaves at most a logarithmic number of strong components.

Part (a) is straightforward by random methods and is given in [6]. Part (b) is given in [10]. This completes the proof. QED

4.2 Bicliques

Calculating the directed toughness of the complete bipartite graph is surprisingly difficult. Assume $2 \leq r \leq s$. Since $\nabla(K(r, s)) = r - 1$, it follows that $\bar{t}(K(r, s)) \leq (r - 1)/(s + 1)$. It remains to find an orientation of $K(r, s)$ with toughness equal to $(r - 1)/(s + 1)$.

We start by investigating a natural orientation D of $K(r, r)$. Say $K(r, r)$ has partite sets U and V with $U = \{u_0, \dots, u_{r-1}\}$ and $V = \{v_0, \dots, v_{r-1}\}$. Then in D there is an arc from u_i to v_j iff $i \leq j \leq i + \lceil r/2 \rceil - 1$. This can be depicted by placing the vertices u_0, v_0, u_1, v_1 , etc clockwise on a circle; then all arcs go clockwise (except for the arcs through center of circle for r odd).

Consider removing a set T of t vertices where $t \leq r - 2$.

Assume first that r is even. We define two vertices as **antipodes** if they are in the same partite set but their out- and in-neighborhoods are reversed. For example, u_i and $u_{i+r/2}$ are antipodes. Every vertex has an antipode. Therefore the removal of $r - 2$ vertices must leave a pair P of antipodes.

CLAIM. *If $P \subseteq U$, then in $D - T$ all the vertices of $V - T$ are in the same strong component. Similarly if $P \subseteq V$.*

Say $P = \{u_0, u_{r/2}\}$. If $D - T$ contains both an in- and an out-neighbor of u_0 , then these two vertices together with P form a directed 4-cycle. Further, any vertex of V has an arc into P and out of P , so the claim is established in this case.

So assume there is no out-neighbor of u_0 . Let v_i denote the first vertex of $V - T$ and v_j the last vertex of $V - T$. Then at most $j - i - 1$ vertices are removed from U . So there exists an antipodal pair $P' = \{a_k, a_{k-r/2}\}$ with $i < k \leq j$. But then we are back in the first case using P' instead of P . This completes the proof of the claim. QED

It follows that $D - T$ has one large component containing all of $V - T$ and some of U , while the remaining vertices of U are sources or sinks. For a vertex to be isolated, all its out-neighbors or all its in-neighbors must be in T . These neighbors are $r/2$ consecutive vertices. Thus the maximum number of isolates is obtained by T being t consecutive vertices of V : this gives $2(t - r/2 + 1)$ isolates (half of the isolates are sinks and half sources) for $t \geq r/2$. (If $t < r/2$ then no isolates and $D - T$ is strongly connected.)

Now we convert this orientation into an orientation of $K(r, s)$ by duplicating each v_i to form a bucket V_i of approximately s/r vertices. It is easily checked that the above claim still holds. So the maximum number of components is obtained by removing either t consecutive vertices of U or t consecutive vertices of V . It is clear that the former gives more components.

In particular, we ensure that $|V_i| = \lfloor (i+1)s/r \rfloor - \lfloor is/r \rfloor$. So the total number of isolates is at most

$$\alpha(t) = 2\lceil (t - r/2 + 1)s/r \rceil.$$

Without the ceilings, the formula $t/(1 + \alpha(t))$ is always minimised at $t = r - 2$, where its value is larger than $(r - 1)/(s + 1)$.

With the ceilings, the ratio is at least

$$t/(2(t - r/2 + 1)s/r + 1 + 2(r - 1)/r)$$

which is minimized at $t = r - 2$. With some calculations (and more careful handling of $s = 2r + O(1)$), it follows that this ratio is at least $(r - 1)/(s + 1)$ for all $s \geq 2r$. (There really is a problem, for example, for s just a bit smaller than $2r$.)

Now assume r is odd. Then we define two vertices as antipodes if they are u_i and $u_{i+\lceil r/2 \rceil}$ or v_i and $v_{i+\lceil r/2 \rceil}$. By a similar argument to the claim above, one can show that there is a strong component which contains all of one partite set except possibly for one vertex z . To isolate the vertices in the other partite set, one must remove all out- or all in-neighbors, except possibly z . Since we are assuming $s \geq 2r$, this cannot occur in V , and so we may assume T is t consecutive vertices of U .

This yields $2((t - r/2 + 1)s/r + O(1))$ isolates, as before. And the minimum thereof is too large for $s \geq 2r + O(1)$. Hence we have established the following theorem:

Theorem 2 *Consider the complete bipartite graph $K(r, s)$ with $s \geq 2r + O(1)$. Then $\bar{t}(K(r, s)) = (r - 1)/(s + 1)$.*

The above formula is an upper bound for all complete bipartite graphs but we do not know if it is always attainable. In at least one case, there is not equality. Consider $K(3, 4)$. Then, for every degree-3 vertex there is a vertex whose removal isolates it. One such vertex must handle at least two degree-3 vertices. That is, there exists a vertex whose removal yields at least 3 strong components. Thus $\tau \leq 1/3$ for any orientation. Nevertheless, we conjecture this is the only exception:

Conjecture 1 *Consider the complete bipartite graph $K(r, s)$ with $2 \leq r \leq s$. Then $\bar{t}(K(r, s)) = (r - 1)/(s + 1)$ except for $r = 3$ and $s = 4$.*

It is to be noted that the above construction does not always do the job. Indeed, $K(4, 6)$ achieves the upper bound, but there is no suitable regular orientation (meaning for each vertex its in-degree equals its out-degree). For many other small cases, a computer has been used to verify the conjecture.

4.3 Cubic Graphs

For cubic graphs G ,

$$\bar{t}(G) \leq \bar{\kappa}(G)/2 \leq 1/2.$$

We know of one cubic graph, $K(3, 3)$, that has directed toughness equal to $1/2$. It is not clear what the maximum value of directed toughness is for large cubic graphs. The best we have is $1/3$ obtained by the prism $K_m \times K_2$ for m even. (The best orientation is such that each 4-cycle is a directed cycle.)

As one piece of evidence, we note that a cubic graph with a triangle has directed toughness at most $1/3$. For, in any strongly connected orientation, there is either exactly one arc into the triangle or exactly one arc out of the triangle. Removing the vertex in the triangle incident with that arc means that the other two vertices in the triangle are each in a singleton strong component.

4.4 Open

There is no intrinsic connection between directed toughness and hamiltonicity: witness the low directed toughness of the cycle. Nevertheless, the

directed toughness is a lower bound for toughness, so a (very?) weak version of Chvátal's conjecture suggests itself: does a sufficiently large value of the directed toughness guarantee hamiltonicity? Possibly even $\bar{t}(G) \geq 1$?

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